

AI/ML intern DAY 19

AI/ML Internship Final Project Documentation

AI-Based Virtual Try-On and Fashion Recommendation System

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1. Introduction

The rapid growth of Artificial Intelligence and Computer Vision has significantly transformed the fashion industry. Modern e-commerce platforms increasingly rely on intelligent systems that help users visualize clothing, receive personalized recommendations, and reduce uncertainty before making a purchase.

One of the most promising applications in this field is the AI-Based Virtual Try-On (VTON) system. Instead of physically wearing garments, users can upload their images and digitally preview clothing items using advanced computer vision algorithms.

This documentation summarizes the research, technical concepts, workflows, and findings explored during the AI/ML Internship. The project focused on understanding the technologies required to build an intelligent Virtual Try-On platform integrated with AI-powered fashion recommendations.

2. Project Goals

The primary goals of the internship project were:

- Understand Computer Vision fundamentals.
- Study human pose estimation techniques.
- Explore body segmentation methods.
- Research body landmark detection.
- Analyze body measurement extraction.
- Investigate AI-based garment warping.
- Study 2D and 3D Virtual Try-On models.
- Explore AI recommendation systems.

- Design a complete AI workflow for a fashion platform.
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3. Project Scope

The project covered multiple AI domains that work together to create a modern fashion technology platform.

Computer Vision

- Human Detection
- Body Segmentation
- Pose Estimation
- Landmark Detection
- Background Removal

Virtual Try-On

- 2D Virtual Try-On
- 3D Avatar Generation
- Garment Warping
- Cloth Alignment
- Image Blending

Artificial Intelligence

- Recommendation Systems
 - Deep Learning Models
 - Geometric Matching
 - Human Reconstruction
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4. Technologies and Tools Studied

Programming Tools

- Python
- OpenCV
- MediaPipe
- TensorFlow
- GitHub
- Visual Studio Code

AI Frameworks

- MediaPipe Pose
- OpenPose
- Detectron2
- DeepLabV3
- U2Net

Virtual Try-On Models

- VITON
- CP-VTON
- HR-VITON
- StableVITON
- IDM-VTON

3D Human Modeling

- SMPL
- PIFuHD
- ICON
- HumanNeRF
- 3D Gaussian Splatting

5. Research Areas Explored

5.1 Human Pose Estimation

Pose estimation identifies important body joints that are required for garment alignment and body measurement estimation.

Models studied:

- MediaPipe Pose
- OpenPose

Applications:

- Virtual Try-On
- Fitness Tracking
- Motion Analysis
- Human Reconstruction

5.2 Body Segmentation

Body segmentation separates the user from the background and identifies important body regions.

Segmentation models explored:

- DeepLabV3
- SCHP
- Detectron2
- MediaPipe Selfie Segmentation

Segmentation improves garment placement and fitting quality.

5.3 Body Landmark Detection

Body landmarks provide reference points for calculating body proportions.

Major landmarks include:

- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles

These landmarks are used for body measurement estimation and garment alignment.

5.4 Body Measurement Extraction

One of the major research areas involved estimating body dimensions from images.

Measurements studied:

- Height
- Shoulder Width
- Waist Width
- Hip Width
- Arm Length

The workflow consists of:

Input Image



Pose Detection



Landmark Extraction



Distance Calculation



Body Measurement Estimation

6. Fashion Datasets

Several publicly available datasets were studied.

Dataset	Primary Application
DeepFashion	Fashion Recognition
DeepFashion2	Landmark Detection
VITON	2D Virtual Try-On
VITON-HD	High Resolution Try-On
Dress Code	Multi-category Clothing
MPV Dataset	Multi-pose Virtual Try-On
Fashionpedia	Fashion Attributes

Dataset Preparation

The following processes were researched:

- Data Collection
- Data Cleaning
- Metadata Generation
- Annotation
- Training/Test Split

7. Garment Warping and Fitting

Garment warping transforms clothing images so they naturally adapt to different body shapes.

Techniques Studied

Thin Plate Spline (TPS)

- Smooth geometric transformation.
- Landmark-based deformation.
- Computationally efficient.

Geometric Matching Module (GMM)

- Deep learning based.
- Learns garment-body correspondences.
- Improves fitting accuracy.

Flow-Based Warping

- Pixel-level movement estimation.
- Better texture preservation.
- Suitable for complex poses.

8. 2D Virtual Try-On Systems

Several Virtual Try-On architectures were analyzed.

Model	Key Feature
VITON	Basic 2D Try-On
CP-VTON	Characteristic Preservation
HR-VITON	High Resolution Output
StableVITON	Diffusion-based Generation
IDM-VTON	Identity Preservation

The study showed that HR-VITON provides better texture quality and garment realism compared to earlier models.

9. 3D Avatar Generation

3D reconstruction technologies allow users to create digital human avatars.

Models researched:

- SMPL
- PIFuHD
- ICON

- HumanNeRF
- Gaussian Splatting

These technologies enable future development of immersive Virtual Try-On experiences.

10. AI Recommendation Systems

Modern fashion platforms use recommendation systems to personalize shopping experiences.

Methods studied:

Content-Based Filtering

Recommends products based on item similarity.

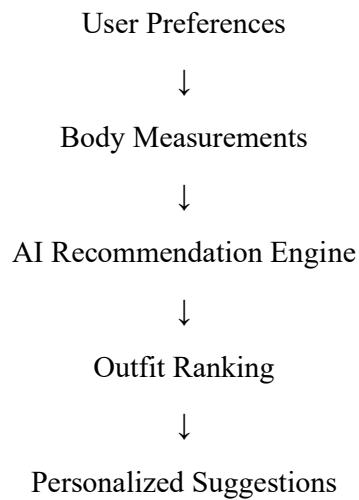
Collaborative Filtering

Uses behaviour patterns from multiple users.

Hybrid Recommendation

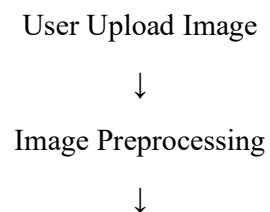
Combines both approaches for improved accuracy.

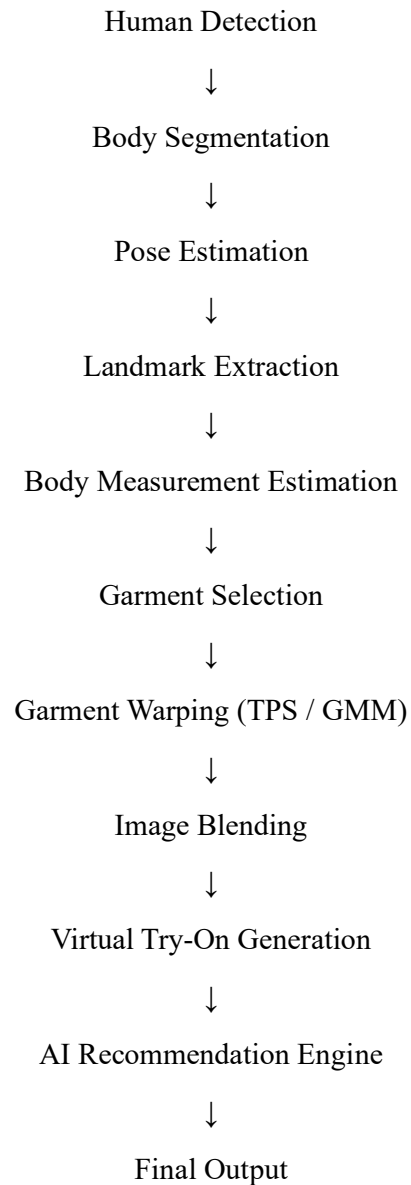
Proposed workflow:



11. Complete AI Workflow

The complete AI pipeline designed during the internship is:





12. Skills Developed

Technical Skills

- Artificial Intelligence
- Computer Vision
- Machine Learning Research
- Pose Estimation
- Body Segmentation
- Virtual Try-On Systems

- 3D Human Reconstruction
- Python Programming
- OpenCV
- GitHub

Professional Skills

- Technical Documentation
 - Research Analysis
 - Comparative Study
 - Problem Solving
 - Independent Learning
 - Project Organization
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13. Challenges Encountered

Several technical and research challenges were identified during the internship.

- Understanding advanced research papers.
 - Learning multiple AI architectures.
 - Comparing numerous Virtual Try-On models.
 - Studying complex 3D reconstruction techniques.
 - Understanding complete fashion AI workflows.
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14. Future Scope

Future improvements for AI-powered fashion platforms include:

- Real-time Virtual Try-On.
 - Mobile application deployment.
 - Full 3D avatar integration.
 - Personalized AI fashion assistants.
 - Multi-angle garment visualization.
 - Diffusion-model based clothing generation.
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15. Conclusion

The AI/ML Internship provided comprehensive exposure to modern Artificial Intelligence applications in the fashion industry. The research covered body segmentation, pose estimation, body measurement extraction, garment warping, 2D and 3D Virtual Try-On systems, recommendation engines, and fashion datasets.

The internship demonstrated how multiple AI technologies work together to build intelligent fashion platforms capable of improving online shopping experiences. The knowledge gained through this project established a strong foundation in Computer Vision, Machine Learning, and AI-driven Virtual Try-On technologies while providing valuable insight into future developments in digital fashion and e-commerce.